

CHAPTER TWO

Computer Cases and Motherboard

- ⦿ Computer Cases
- ⦿ Mother board
- ⦿ Safety

Computer Cases

Computer Cases

- ☑ Provides protection and support for internal components
- ☑ Should be **durable, easy to service**, and have enough **room for expansion**
- ☑ Contain the framework to support and enclose internal components of the computer
- ☑ Typically made of plastic, steel, and aluminum
- ☑ Available in a variety of styles

Computer Cases



- ☑ The size and layout of a case is called a **form factor**
- ☑ Computer cases are referred to in a number of ways:
 - ☑ Computer chassis
 - ☑ Cabinet
 - ☑ Tower
 - ☑ Box
 - ☑ Housing

Case Selection

Factor	Rationale
Model Type	• Two main case models (one for desktop PCs and the other for tower PCs). • The type of motherboard determines the type of case. • Size and shape must match exactly.
Size	If a computer has many components, it will need more room for airflow to keep the system cool.
Available Space	• Desktop cases allow space conservation in tight areas because the monitor can be placed on top of the unit. • The case design may limit the number and size of the components that can be added.
Power Supply	Match the power rating and connection type of the power supply to the type of motherboard chosen.
Appearance	There are many case designs to choose from if it is necessary to have a case that is attractive .
Status Display	LED indicators that are mounted on the front of the case can tell you : • if the system is receiving power, • when the hard drive is being used, and • when the computer is on standby or sleeping.
Vents	• All cases have a vent on the power supply. • Some cases have more vents to dissipate an unusual amount of heat.

Purpose of Computer case

- ☑ The real value of a case does not lie in the label, color, or how pretty it looks.
- ☑ Computer cases have the purpose of:
 - ☑ **house** all the internal components
 - ☑ keeps internal components **cool**
 - ☑ offers **access** to the external world through ports and connectors
 - ☑ **prevent** delicate circuits from **damage**
 - ☑ **prevent EMI**(Electromagnetic Interference)from pc from affecting surrounding devices such as TV, telephone, etc

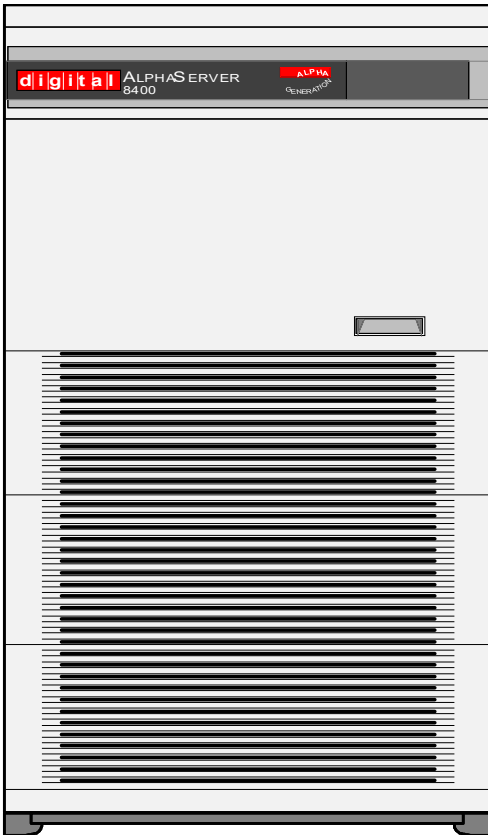
Computer cases ...

- ☑ EMI-is a radio frequency that is emitted from electronic or electrical device.
- ☑ It interferes with surrounding equipment.
- ☑ A computer interferes with radio, telephone, TV, etc when it generates EMI.
- ☑ Running a computer without cases causes interference to surrounding devices.
- ☑ *It is not a good idea to run a computer for extended periods of time with the case open or removed entirely.*
- ☑ This not only produces EMI, but also results in improper airflow and reduced cooling of the system components.

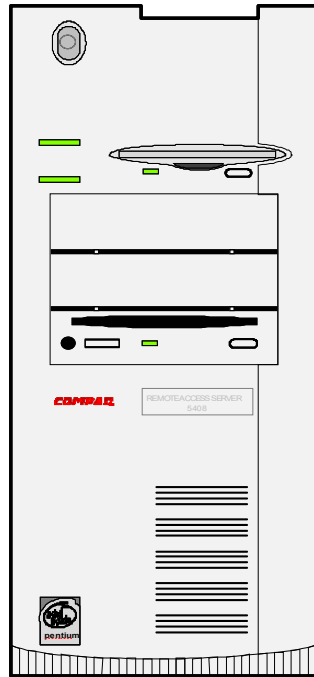
Types of cases

- ☑ There are different types of cases:
 - ☑ Full-sized tower
 - ☑ Mid-size tower
 - ☑ Mini-tower
 - ☑ Desktop
 - ☑ Slim desktop
- ☑ Slim desktops integrate the display into housing.
- ☑ The monitor and system unit are combined into one unit.
E.g. imac G5/

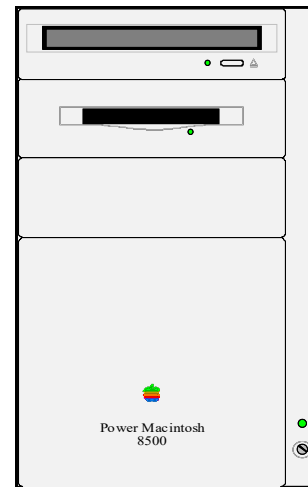
Cases



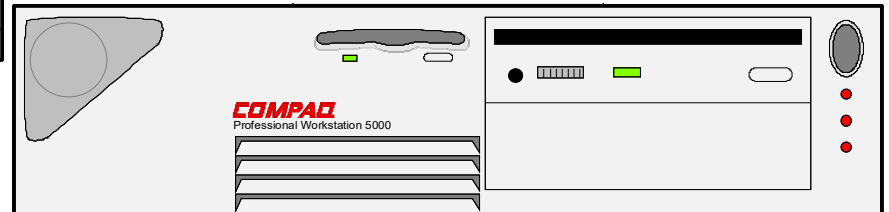
Full-size tower



Mid-size tower



Mini-tower



Desktop

Cases



micro, mini, mid, full tower, & desktop cases

Case ...

- ☑ When we consider a case, the following things are cared about:
 - ☑ The **bigger** the box:
 - ☑ the more components it can **hold**
 - ☑ This gives it more **expansion capability**
 - ☑ **better air flow** which is good for cooling.
 - ☑ Large cases are **easier to work** with.
 - ☑ The **more compact** the box:
 - ☑ the **less expansion** potential it has.
 - ☑ Working with such cases is **difficult**
 - ☑ usually **air flow is reduced**.

Safety Measure

- ☑ The human body has an electric field and under the right conditions can generate a tremendous amount of voltage, often referred to as static electricity.
- ☑ When we open a computer case, we need to make sure that **Electro Static Discharge(ESD)** does not affect the computers' internal components.

Safety measures

- ☑ ESD occurs when an imbalance in the amounts of positive and negative charges on the surface of an object is released.
- ☑ The most dramatic example of ESD is lightning, which splits trees as easily as it lights up the sky.
- ☑ The amount of energy released when you touch a metal object can be quite large.
- ☑ The buildup of energy with nylon clothes (wool is less) can easily reach 21,000 volts.
- ☑ About 750 volts are required to produce a visible spark with ESD, while a mere 10 volts or so can ruin a computer chip.

Safety measures

- ☑ Sources of ESD are:
 - ☑ The **human body** has an electric field and it can generate static electricity.
 - ☑ Another source of ESD is **dust** that build upon pc components. The dust can create charges that could damage internal components of PC.
- ☑ To avoid this, we have to clean the PC regularly. **Blower** is the best tool for this.

ESD Damage

- ☑ The amount of damage caused by ESD can be divided into three categories:

1) Catastrophic failure:

- ☑ This damage causes a computer component to fail immediately.
- ☑ Heat, sometimes noise and smoke are generated during the process.
- ☑ This type of failure is easy to identify as the device fail completely.
- ☑ You can replace such components.

2) Upset failure (degradation failure):

- ☑ ESD causes erratic(unpredictable)fault in a component.
- ☑ This problem is very difficult to detect and repair because the failure is intermittent(irregular).
- ☑ The best way to solve this problem is replacing suspected parts until the fault stops appearing.

3) Latent failure:

- ☑ this type of failure weakens the actual transistor.
- ☑ The part will seem normal in most operations
- ☑ will frequently pass quality control and conformance tests.
- ☑ Like upset failure, it is difficult to identify.

Preventing ESD

- ☑ Preventing ESD is the best defense against it.
- ☑ The number one cause of ESD is **improper handling of electronic devices** during installation or maintenance.
- ☑ The key prevention to ESD is to keep all electronic components and yourself at a common electrical point.
- ☑ When you open a computer, **ground yourself by touching the case (chassis) of a computer.**

Preventing ESD

- ☑ There are also devices that enable us to protect ESD. These devices include:

1) **Anti-static wristband:**

- ☑ these devices have a strap that you wrap around your wrist or ankle on one end.
- ☑ The other end of the strap is **attached computer case** to keep ground between the computer and you.

Anti-static wristband



Preventing ESD

- 2) **Anti-static mats:** these are mats that you place on the work surface and on the floor in front of work area. They make an excellent place to place your tools and components when you work on a system.



Fig :Anti-static mat

- 3) **Anti-static bags (pouches):**

- ✓ they have special coating or filament that prevents ESD.
- ✓ They are used to store any sensitive electronic device like cards, RAM, drives, etc when they are not installed in a computer.

- 4) **Anti-static pad:**

- ✓ It is an insulating foam pad in which individual chips with exposed pins should be embedded when they are not installed in a computer.

Rules for opening computer:

1. Turn off (shut down) the computer
2. Remove the power cable (unplug it)
3. Remove peripheral devices like mouse, printer, keyboard, etc.
4. Open the case. Take care of ESD.
 - ☑ Wear anti-static wristband or hold the case to ground yourself.
5. Take care when you hold internal components of computer like RAM, Cards, CPU, etc.
 - ☑ Hold them on the sideways.

Motherboard

- ☑ The **main circuit board** that contains most of the electronics
- ☑ all computer components **connect to, or communicate through**, the motherboard.
- ☑ Motherboard is the PCs **center of activity**.
- ☑ It defines the computer's **limit of speed, memory, and expandability**.

Chipsets

- ☑ A motherboard comes with a variety of support chips soldered in place.
- ☑ **These chipsets are designed to work with CPU.**
- ☑ These chips are **highly complex and coordinated ICs** that **help CPU manage and control the computer system.**
- ☑ When replacing CPU, you must make sure the CPU works (compatible) with the chipset supported by the motherboard. Otherwise the computer will not work.

Motherboard

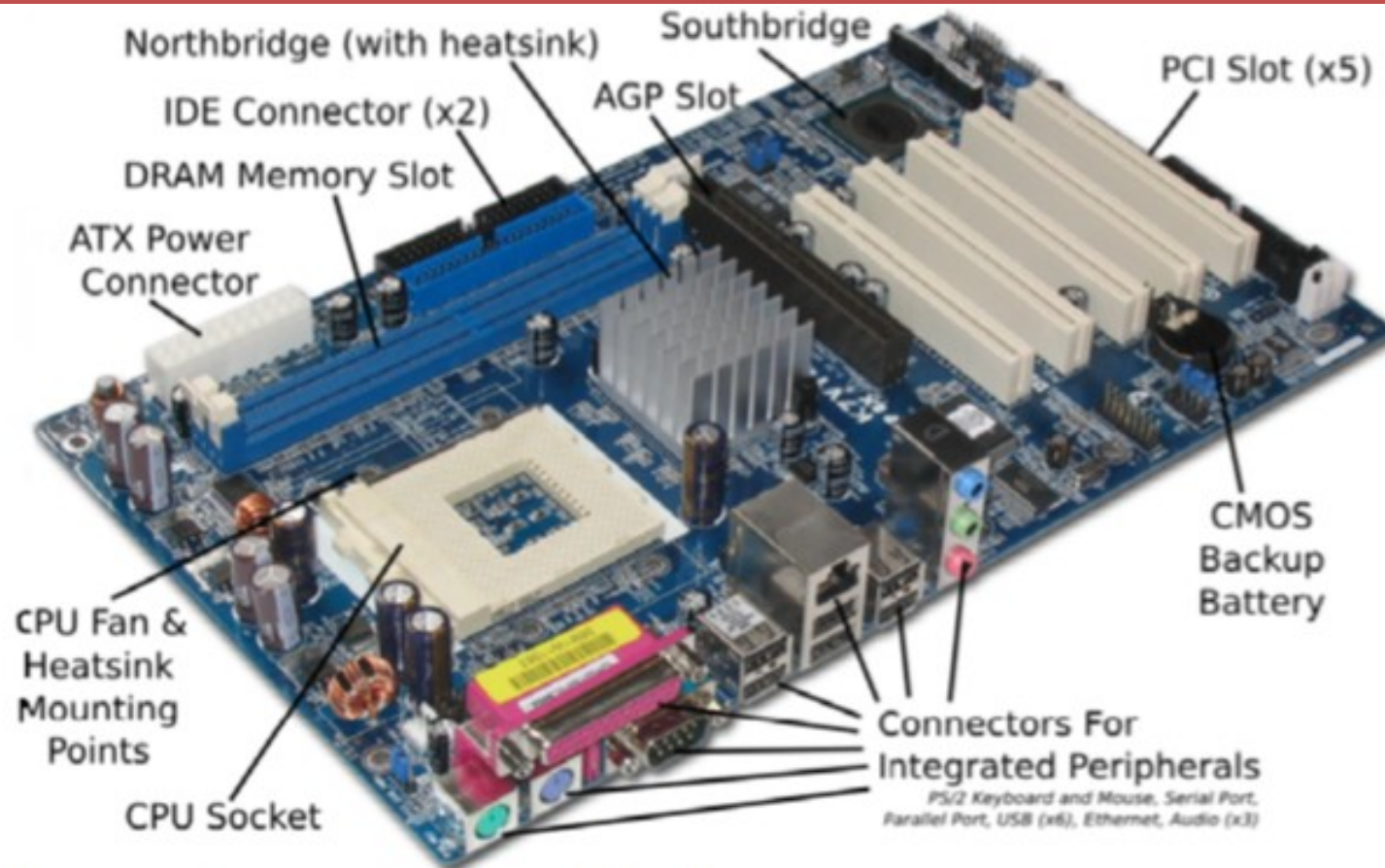
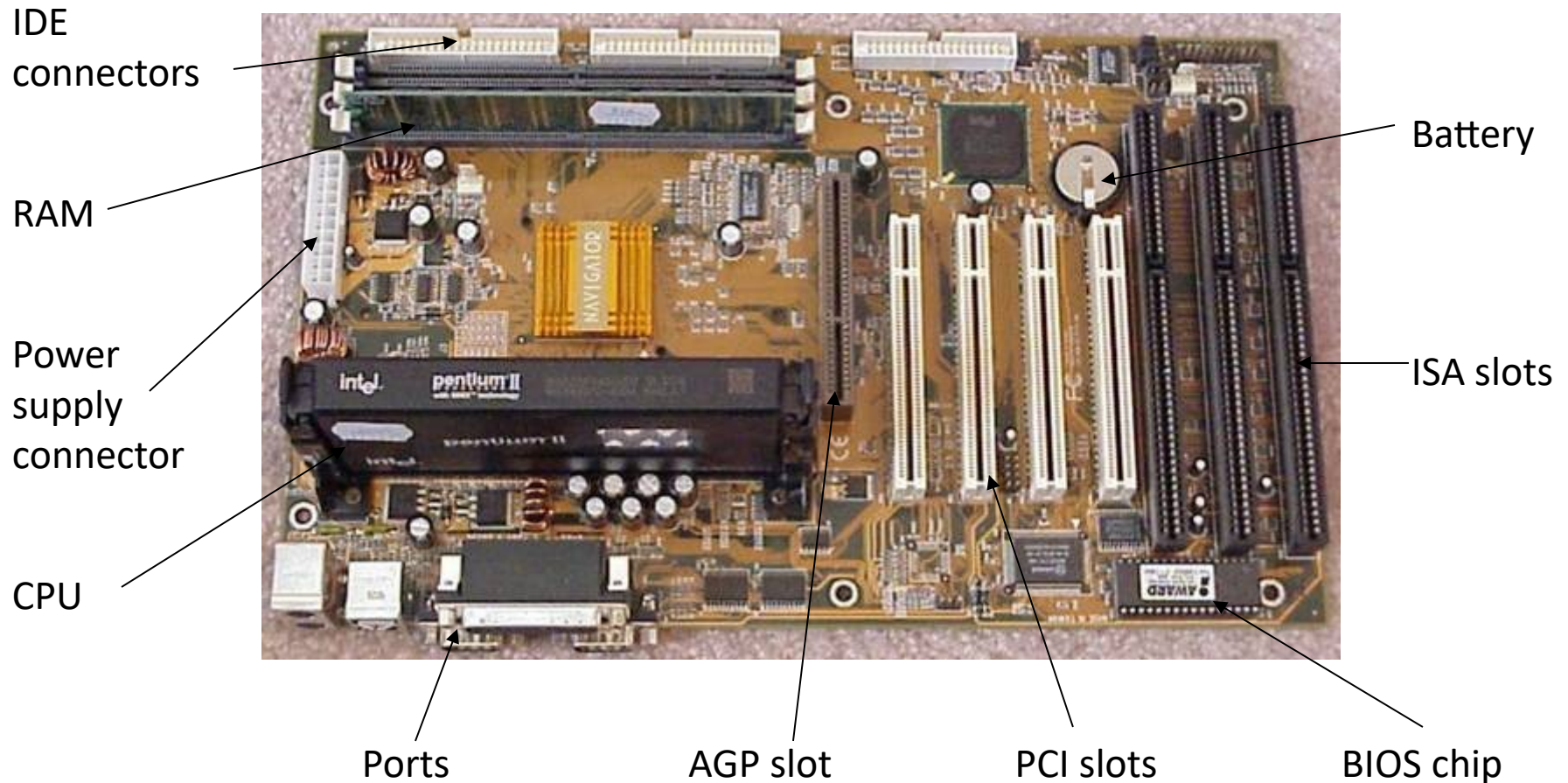


Fig: typical components on the motherboard

Key Components: Motherboard



Chipset

- ☑ A basic chipset consists of:
 - 1) **Bus controller**
 - 2) **Memory controller**
 - 3) **Data and address buffer**
 - 4) **Peripheral controller**
- ☑ **The number of chips a motherboard differs from one type of motherboard to another.**

Chipsets ...

- ☑ New chips often enable:
 - ☑ Higher speed on one or more buses
 - ☑ Utilization of new facilities like new RAM types, new buses, etc.
- ☑ There are several suppliers of Pentium chipsets:
 - ☑ Intel
 - ☑ SIS(Silicon Integrated System)
 - ☑ Via
 - ☑ Ali (Acer Laboratories, inc)

Types of Motherboard

- ☑ Motherboards come in several standardized configurations called **form-factors**.
- ☑ **Form- factors** define
 - ☑ **Speed**
 - ☑ **Capacity**
 - ☑ **CPU type used**
 - ☑ **The size of the motherboard.**
 - ☑ **Location of expansion slot**
 - ☑ **What sorts of power supply it uses**
 - ☑ **Its physical organization Etc**

- ❑ PC motherboard components differ depending on the manufacturer and model.
- ❑ However, all have the following primary features:
 - ❑ BIOS Chip
 - ❑ CMOS Battery
 - ❑ Form Factor
 - ❑ Type of Processor They Support
 - ❑ Chipset
 - ❑ Type of Bus & Bus Speed
 - ❑ Expansion Slots
 - ❑ Memory Slots

Common form-factors

1) Obsolete/outdated/ form factors:

- ☑ Baby AT
- ☑ Full-size AT
- ☑ LPX

2) Modern form factors:

- ☑ ATX
- ☑ Micro-ATX
- ☑ Flex ATX
- ☑ NLX
- ☑ WTX
- ☑ BTX

3) Proprietary designs. E.g. Compaq

Compare Motherboards



Laptop Motherboard



Desktop Motherboard

Pros and Cons of Integrated Motherboards

Pros

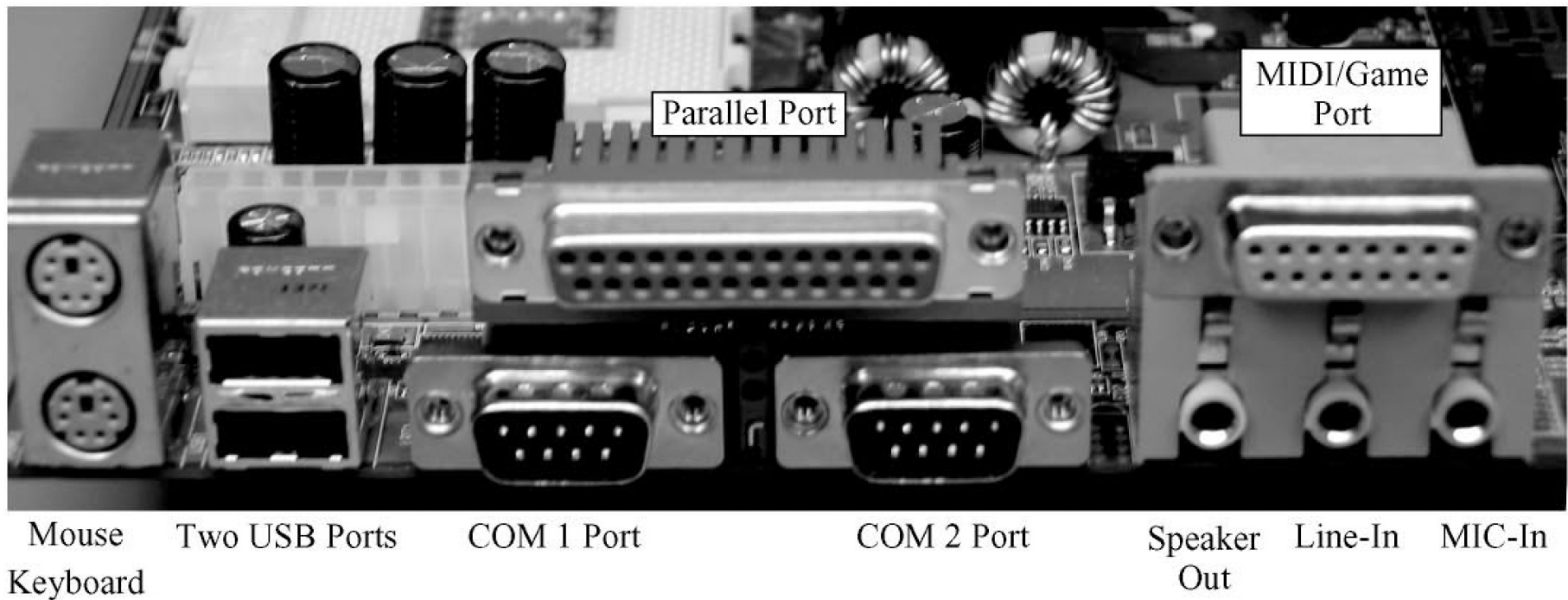
- ☑ Provides expandability because ports are built in and allow more room for expansion cards.
- ☑ Built in ports are **faster** than those on adapters.
- ☑ Computers are **easier to set up** because the manufacturer configures the ports.
- ☑ Systems with integrated motherboards are normally **easier to troubleshoot**.
(few components)

Cons

- ☑ **Motherboard must be replaced when one port goes bad**, unless it supports disabling the faulty port.
- ☑ If ports cannot be disabled, there could be **conflicts with adding expansion cards**.

Pros and Cons of Integrated Motherboards

Integrated Motherboard Ports



Chipset Evolution

- ❑ When the first motherboard was introduced by IBM, they used several discrete chips:
 - ❑ Math coprocessor
 - ❑ Clock generator
 - ❑ Bus controller
 - ❑ System timer
 - ❑ Interrupt and DMA controllers
 - ❑ CMOS RAM and clock
 - ❑ Keyboard Controller

Chipsets ...

- ☑ In 1986, a company called **Chips and Technologies** introduced revolutionary component called the **82C206**.
- ☑ This was a **single chip** that integrated the functions of **Clock Generator, Bus Controller, System Timer, Dual Interrupt Controllers, dual DMA Controllers, and the CMOS/Clock Chip**.
- ☑ Virtually all the major chips on the motherboard could now be replaced by a single chip. This idea was copied by other manufacturers.

Intel Chipsets

- ✓ Intel joined the chipset market in 1989 by introducing the 420 series chipset along with 486 processor.
- ✓ Intel produced CPU before 1989.
- ✓ For new CPUs produced by Intel to be on market, compatible chips are necessary.
- ✓ Before chipsets are produced for new Intel CPU it takes as long as a year.
- ✓ This means Intel could not sell its CPU as needed.
- ✓ Because of this, Intel started to produce chipset for its CPU.

Chipsets...

- ☑ Intel started a pattern of number its chipsets:

Chipset number	Processor family
420XX	P4(486)
430XX	P5(Pentium)
440XX	P6 (Pentium pro/II/III
450XX	P6 server(Pentium pro/PII/PIII Xeon)
8XX	P6(PII/III/IV) hub architecture
9XX	P7(Pentium IV) hub architecture

- ☑ Intel used two distinct chipset architectures:
 - ☑ North/south bridge architecture
 - ☑ Hub architecture

Intel Chipset Architecture

1) North-South Bridge Architecture

- ☑ Most of earlier Intel chipsets and all non-Intel chipsets are broken down into a multi-tiered architecture.
- ☑ This includes:
 - ☑ North bridge
 - ☑ South bridge
 - ☑ Super I/O chip

Intel Chipset Architecture

a) North Bridge:

- ☑ It handles communication between the **CPU, RAM, AGP on PCI Express and South Bridge.**
- ☑ It handles communication with faster devices.
 - ☑ Because different processors and RAM requires different signaling,
- ☑ North Bridge will typically work with:
 - ☑ one or two classes of CPU
 - ☑ and generally one type of RAM.
- ☑ There are a few chipsets that support two types of RAM.
 - ☑ This occurs when there is a shift to new standard.
 - ☑ Because the south bridge is further removed from CPU,

b) South Bridge:

- ☑ It is given the responsibility for slower devices on a typical micro computer.
- ☑ **A particular South Bridge will usually work with different North Bridges**
 - ☑ but these two chips must be designed to work together.
- ☑ connects to the processor indirectly via the north-bridge and controls slower devices such as the hard drive, USB, audio, video, LAN, and PCI.

Intel Chipset Architecture

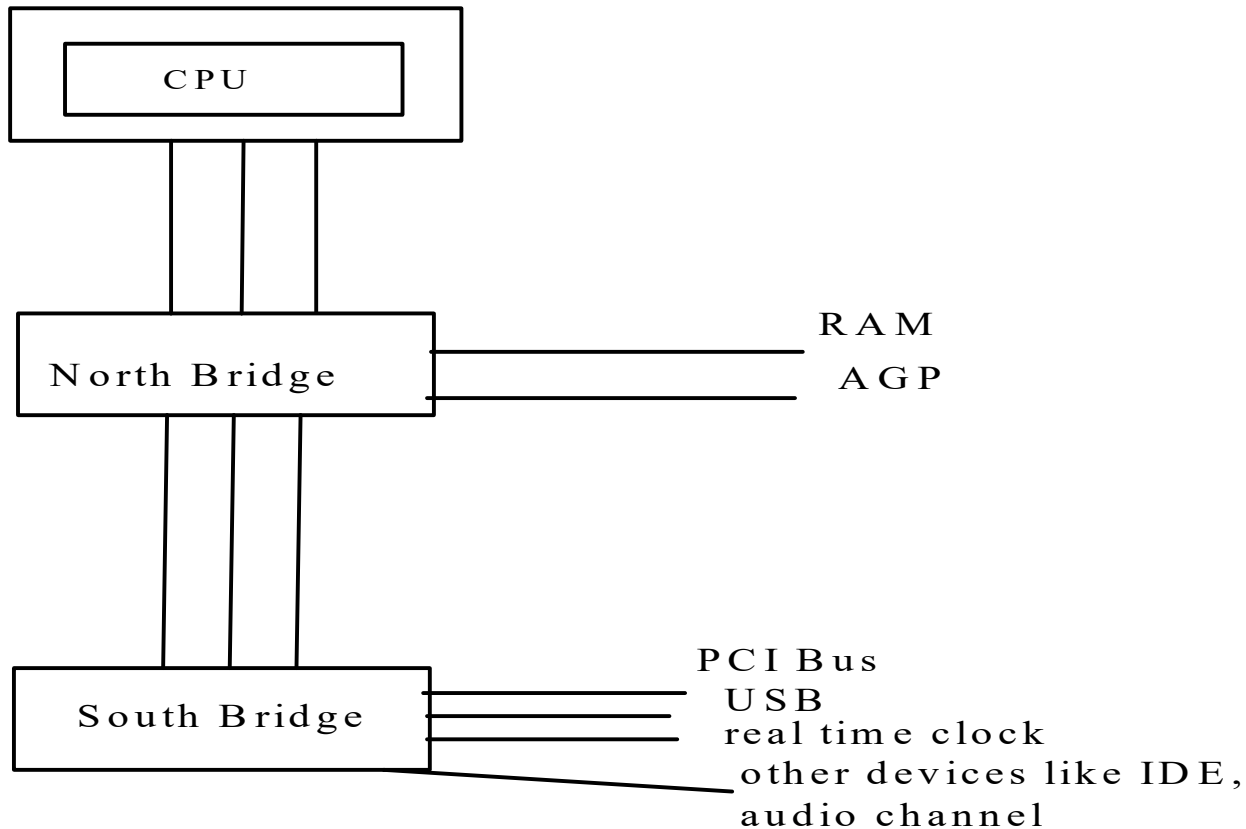


Fig: north and south bridge

Chipsets...

- ❑ Traditionally the interface between North Bridge and South Bridge was the PCI Bus.
- ❑ Most current chipsets use a different interface with higher performance.
- ❑ The functionality found on contemporary(modern) South Bridge includes:
 - ❑ PCI(Peripheral component interconnect) bus
 - ❑ ISA bus(industry standard architecture)
 - ❑ Interrupt controller
 - ❑ IDE controller(SATA or PATA)
 - ❑ Real time clock

Chipsets...

c)Super I/O Chipset:

☑ The major function of Super I/O chips are:

a) Serial Port controller:

☑ super I/O chip controls the serial ports

b) Parallel port control:

☑ it provides the circuitry to drive the parallel port.

c) Floppy disk control:

☑ support for floppy disk drives is provided by the super I/O chip

d) Keyboard and mouse controller

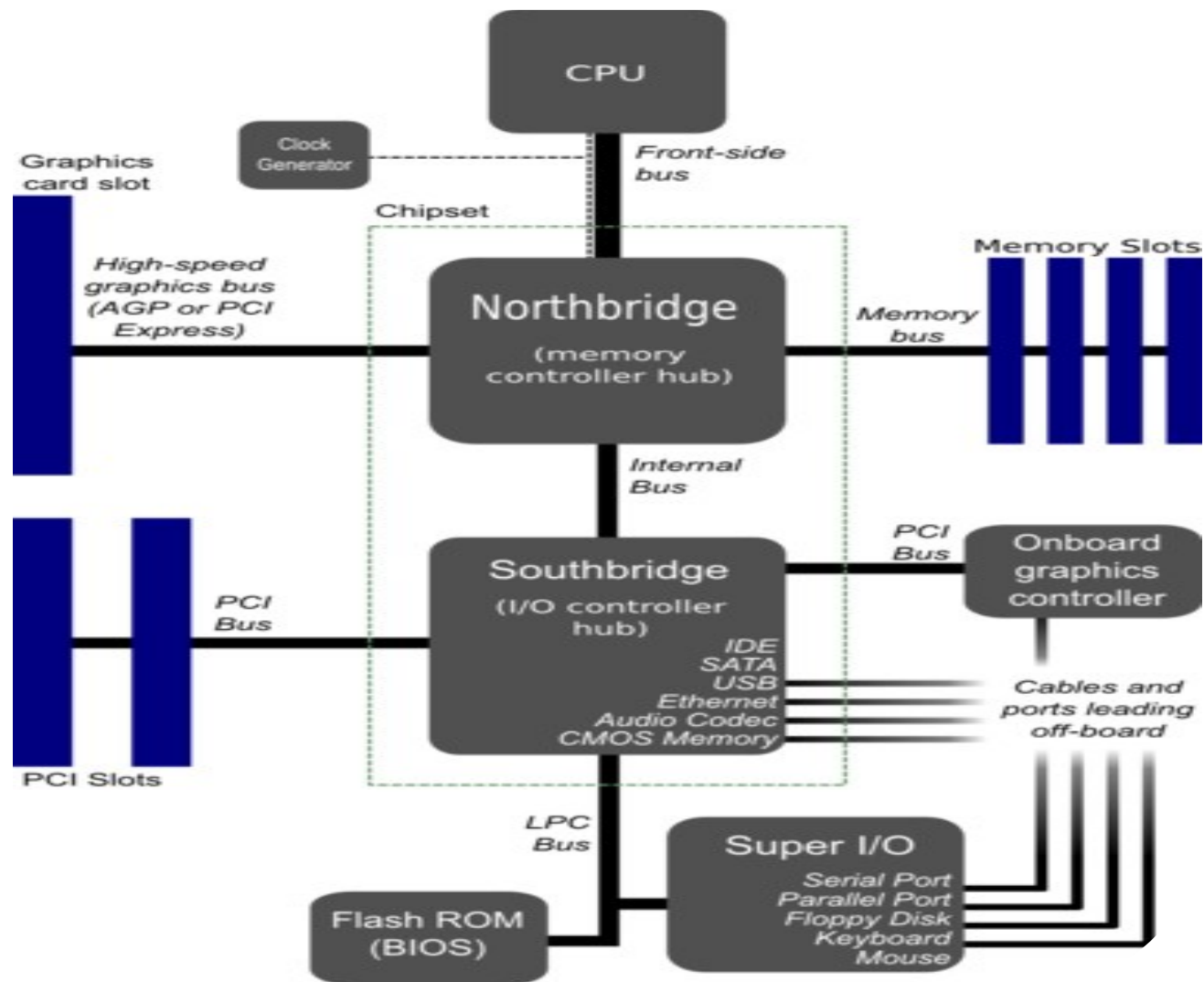


Fig: motherboard chipsets and their interconnection

Chipsets...

2) Hub Architecture

- ☑ The newer 800 and 900 series chip uses a hub architecture
 - ☑ the former **north** bridge is now called **Memory Controller Hub**
 - ☑ the former **South** Bridge is called **I/O Controller Hub(ICH)**.
- ☑ They are **connected** via **4X66MHZ** hub interface which is twice as fast as **PCI** and does not **share** or steal PCI bus **bandwidth**.
- ☑ This improves performance of all PCI connected devices because the PCI is not involved in these transactions.

Hub interface:

- ☑ Allows for much greater throughput for PCI devices
- ☑ Allows greater throughput for devices directly connected to the I/O controller hub.

Chipset Generation

Fifth Generation Chipset (P5):

- ☑ Intel introduced a chipset called 430LX with the introduction of its **first Pentium** processor.
- ☑ Following that, Intel released different versions of this chipset at different times.
- ☑ Some are: 430LX (Mercury), 430NX (Neptune), 430 FX (Triton), 430MX, 430 HX, 430 VX, and 430 TX.

Sixth Generation Chipset (P6):

- ☑ Sixth generation chipsets are chipsets that work with Pentium pro/II/III.
- ☑ The chipsets are of two architectures:
 - ☑ hub architecture and north and south bridge architecture.

Chipsets...

Pentium Pro chipsets are:

- ☑ North and south bridge: 450KX (Orion), 450 GX(Orion server), 440FX(Natoma).
- ☑ Hub architecture: 810(Whitney), 810E (Whitney), 820(Camino), 820E, 840

Intel 810:

- ☑ Introduced in 1999. This chipset represents a major change in chipset design from standard **North South Bridge** to **hub** architecture design. It allows for improvement in system performance all for less cost and system complexity.

Seventh Generation Chipset (P7):

- ☑ With the introduction of P IV, Intel introduced new chipset numbered as 9XX. These include: 910GML, 910GL, 915P, 925X, 925XE, 955X, 975X,etc. All of them use **hub** architecture design.

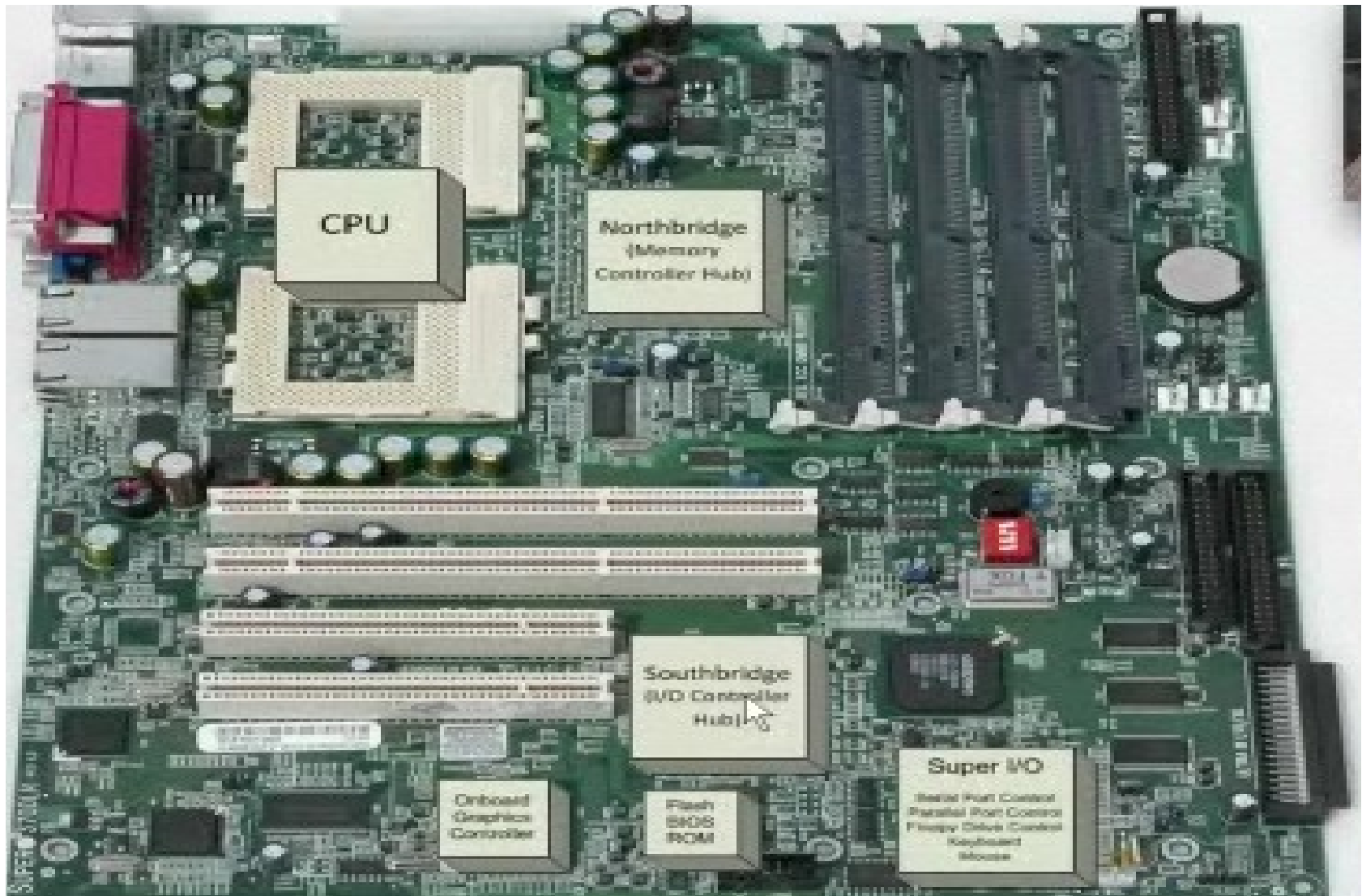


Fig: the motherboard and chipsets

BIOS Chip

- ☑ Every motherboard has a BIOS (Basic Input Output System).
- ☑ This chip contains a small amount of **code** that is **responsible for booting up your system**.
- ☑ It **reads the CMOS settings**, performs the **POST** (Power On Self Test), checks for devices, and then **reads the master boot record** on the hard drive so the operating system can load.
- ☑ **Phoenix** Technologies and American **Megatrends** are the two main BIOS makers.

CMOS Battery

- ☑ Every motherboard has a battery to maintain the time and CMOS settings after the power is turned off.
- ☑ Most are small, round, flat and fit into a socket.

Demo

[Video\BIOS, CMOS, UEFI - What's the difference .mp4](#)



Motherboard Configuration

- ❑ Motherboard settings can be configured in three ways:
 - ❑ Using jumpers
 - ❑ Using DIP switches
 - ❑ a tiny switch or group of switches on a circuit board.
 - ❑ Named for the form factor of the carrier device in which the switch is housed.
 - ❑ Using BIOS

Jumpers

- ☑ Jumpers are **pins** on a motherboard or other device that are used to provide configuration information to the hardware.
- ☑ A single jumper consists of a pair of pins, with a small rectangular shunt that can be placed over both pins to short them together.

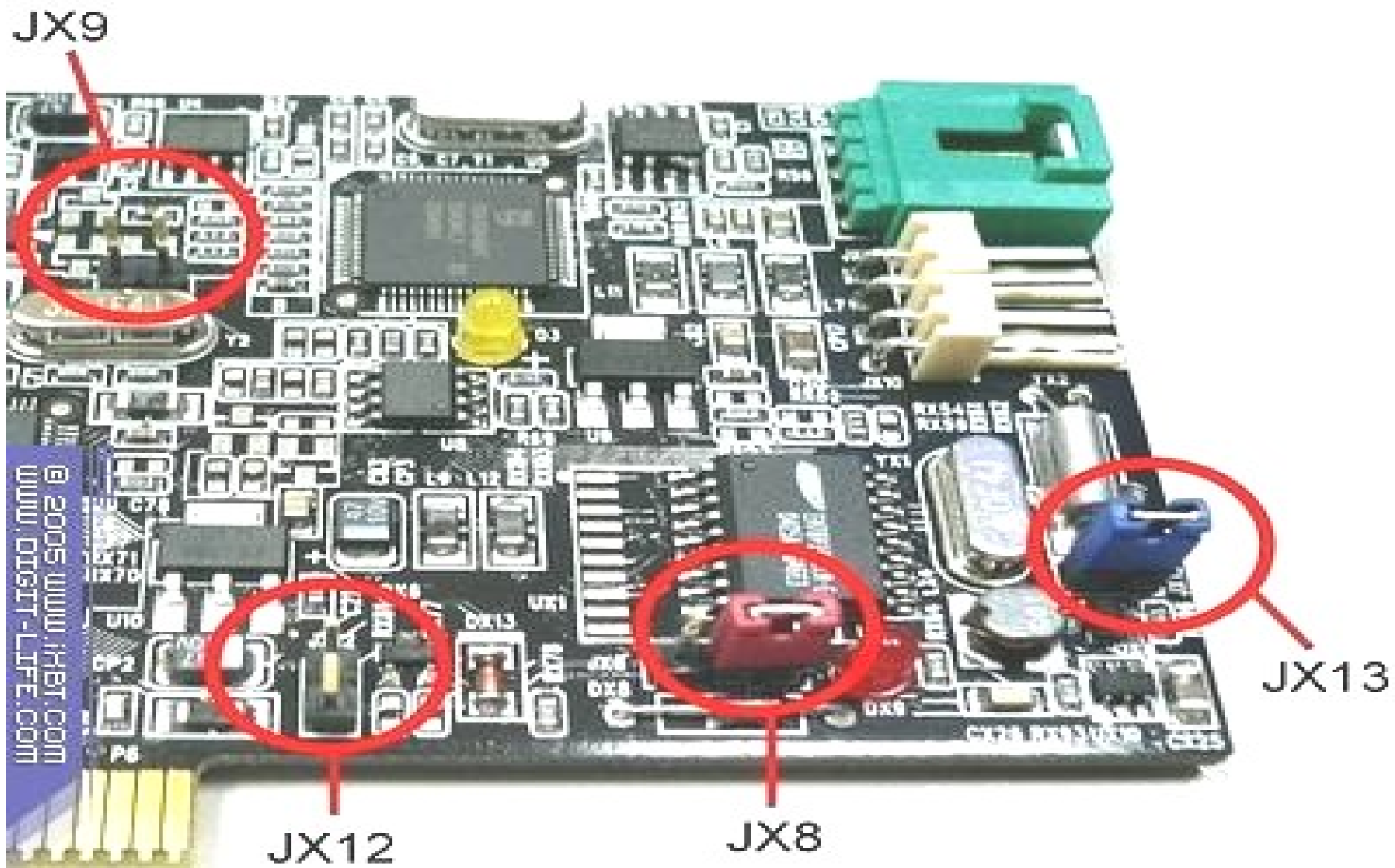


Fig: Jumpers

Jumpers

- ☑ Every motherboard differs in its jumper
 - ☑ numbering,
 - ☑ positioning,
 - ☑ and most importantly, what the settings for each jumper mean.
 - This is why having the motherboard manual is so crucial for anyone who wants to work on their PC.

"jumper less" motherboard

- ☑ the newest innovations on the market
- ☑ With these boards you make many of the hardware settings, such as **CPU type** and **speed**, and even **CPU voltage**, using BIOS settings, while some others are retained as regular jumpers (typically, CMOS clear and cache size).
- ☑ This arrangement lets you change the clock speed of the CPU as easily as you would change any other BIOS setting, and makes upgrading the processor a snap.
- ☑ It also makes it easier for motherboard manufacturers to add support for new processor types as they are put onto the market.

Here is a list of the settings that you are most likely to set using jumpers:

- 1) Processor/CPU Voltage
- 2) Processor Speed / Bus Speed / Multiplier
- 3) Processor Type
- 4) Cache Size and Type
- 5) Memory Size/Type
- 6) Flash BIOS Enable
- 7) CMOS Clear
- 8) Battery Source

Jumpers...

- ☑ Here is a list of the settings that you are most likely to set using jumpers:

1) Processor/CPU Voltage:

- ☑ Almost all newer boards have one or more jumpers to set the **voltage** for the processor.
- ☑ Obviously, this needs to be set correctly or you risk **destroying** the chip.
- ☑ In addition, newer processors use two voltages:
 - ☑ an "external" I/O voltage and an "internal" core voltage.
- ☑ For your motherboard to support these newer processors, it must have jumpers to set both of these voltage types.

Jumpers...

2) Processor Speed / Bus Speed / Multiplier:

- ☑ Every 486 class or later motherboard (excluding jumper less designs) has a way to specify the speed of the processor.
- ☑ There are methods for selecting processor speed, and your manual will show you which your motherboard uses.
- ☑ Some motherboards provide a list of the various speeds supported, and a diagram of how to set the jumpers to match that speed.
- ☑ Others instead require you to set two separate jumpers:
 - ☑ one controls the memory bus speed, and
 - ☑ the other sets the processor "multiplier" (how many times the memory bus speed the processor runs).

Jumpers...

3) Processor Type:

- ☑ In an effort to save the user the hassle of figuring out voltages and speeds, some motherboards combine the jumpers into one jumper set or group.
- ☑ Then, they provide a long list with each processor type and speed that the board supports, and tell you how to set each of a group of jumpers so that it works appropriately.
- ☑ If you are setting up a Pentium 133, you'll find it on the list and be told how to set jumpers 1, 3, 7, 12, and 18, for example.
- ☑ The only problem with this arrangement is that unless they also tell you what each of the individual jumpers is actually controlling, it can be difficult to set up a newer processor that may be supported just fine by the board but not be listed in their documentation.

Jumpers...

4) Cache Size and Type:

- ☑ Some boards can have different amounts of cache, and some can have cache either built in.
- ☑ There is often a jumper or two to set which is used and how much is on the board.

5) Memory Size/Type:

- ☑ Almost all newer PCs auto-detect the amount and type of RAM in the system,
- ☑ but many older 486 (and earlier) models require jumpers to be changed when changing the amount of memory.

Jumpers...

6) Flash BIOS Enable:

- ☑ Many boards require you to set a jumper to a special position in order to enable the flash BIOS update feature.

7) CMOS Clear:

- ☑ Some newer PCs have built into the motherboard the ability to clear the BIOS by temporarily setting a jumper and then replacing it (in essence this disconnects the CMOS memory from the battery so it discharges and resets).

Jumpers...

8) Battery Source:

- ☑ Some boards let you switch the battery from an internal to an external source, and a jumper is used to control the setting.

Jumpers...

Configuration Procedure

- ☑ This procedure describes the steps required to configure a motherboard for installation.
- ☑ You will normally use this procedure when building a new PC or performing a motherboard upgrade.
- ☑ Configuring the motherboard is traditionally done using jumpers.

Steps:

- 1) Read motherboard manual
- 2) Set Processor Voltage Jumper(s)
- 3) Set Processor Speed Jumpers

Jumpers...

4) Set Secondary Cache Size and Type Jumper:

- ☑ Some motherboards accept cache in different configurations and therefore have jumpers depending on how much cache is on the board and whether it is soldered on or inserted via a cache module.

5) Check Flash BIOS Jumper:

- ☑ If your system has a jumper to enable the flash BIOS feature, check it to make sure that it is disabled.
- ☑ This should be the default.

Jumpers...

6) Check CMOS Clear Jumper:

- ☑ Some systems have a jumper that will let you clear the contents of the CMOS memory, something that is needed most often when a system password is set and then forgotten.
- ☑ Make sure that this jumper is set to the normal or default position, or you won't be able to set any BIOS settings.

Jumpers...

7) Check Battery Source Jumper:

- ☑ Some motherboards use a jumper to determine if the onboard battery is to power the CMOS memory, or an external battery.
- ☑ Again, make sure this is set to the default position (onboard battery) unless using an external.

8) Check Disable Jumpers:

- ☑ Some motherboards have special jumpers to allow you to enable or disable parts of the motherboard at a hardware level
- ☑ (for example, the serial/parallel ports or the floppy disk controller). Make sure that these jumpers are set properly (normally, you won't want to disable any of these items).

Jumpers...

9) Set Memory Size Jumpers:

- ☑ Very rarely seen on new boards, jumpers to set the size of the system memory were common on 486-class boards.
- ☑ Set these if your board requires them.



Buying Motherboard: Considerations

- ☑ **Form factor:**
 - ☑ will it fit in my case?
 - ☑ Will it work with the power supply I have selected?
- ☑ **FSB(front-side bus) speed:** faster is better
- ☑ **Chipset:** who makes the Northbridge and Southbridge?
- ☑ **CPU socket:**
 - ☑ will my CPU fit into the board?
 - ☑ Did I select the correct CPU type for my board?
- ☑ **RAM type:**
 - ☑ will my existing RAM type work?
 - ☑ Did I select the correct RAM type?
- ☑ **Expansion:**
 - ☑ are there extra ports and slots to add additional components later?

Motherboard failures

Motherboard failures can be divided into three types:

- ☑ Power failure
- ☑ Bus failure
- ☑ Component failure.

Reasons a motherboard fails

A Motherboard can fail for a number of reasons:

- ☑ Electricity and voltage spikes
- ☑ Dust, smoke
- ☑ Age
- ☑ Fan failure or heating issues
- ☑ Force or pressure

Troubleshooting Motherboard

Here are lists of common problems of motherboard:

1) When you shutdown the PC, it loses date, time, and BIOS settings.

☑ **Solution:** Change CMOS battery. It is exhausted.

Troubleshooting Motherboard

2) Your mouse or keyboard does not work.

- ☑ These ports are not built to withstand repeated insertion.
- ☑ The solution is to resolder them or have professional resolder fix them.

Troubleshooting Motherboard

3) The system does not work after installing new components:

☑ **Solutions:**

- ☑ Check if the CPU, RAM, or AGP(accelerated graphics port) card is properly installed or fit
- ☑ Remove the new component and run the system
- ☑ Check the system for some sort of burn during the process
- ☑ Check the jumper setting of the system.

Tips to troubleshoot motherboard:

- 1) Remove peripheral components from motherboard including expansion cards, external peripherals, etc. and see if the problem is solved
- 2) Check the motherboard jumper setting: make sure that they are all correct and not causing the problem.

Tips to troubleshoot motherboard:

3) Reset BIOS setting to default and check

4) Inspect the motherboard for:

- ☑ Is Motherboard cracked?
- ☑ Is there any broken pin?
- ☑ Is power Connector fit properly?
- ☑ Is there any loose socketed components?
- ☑ Are all cables connected properly?

Videos

- ✓ [Video\1Major parts & components of the Motherboard.mp4](#)
- ✓ [Video\2Parts of the Motherboards and its Functions \(HD\).mp4](#)

END OF THE CHAPTER

Thank you